Quantization Error Analysis

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Key Concepts

1) Quantization error is usually modeled as a random process and characterized statistically.

2) Usual model for quantization error:

    a) The error is stationary, that is, the statistical properties do not change over time.

    b) The error is uniformly distributed between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$ where Δ is the quantization step size.

c) The error is uncorrelated with itself, that is, values of the error at any two times appear unrelated.

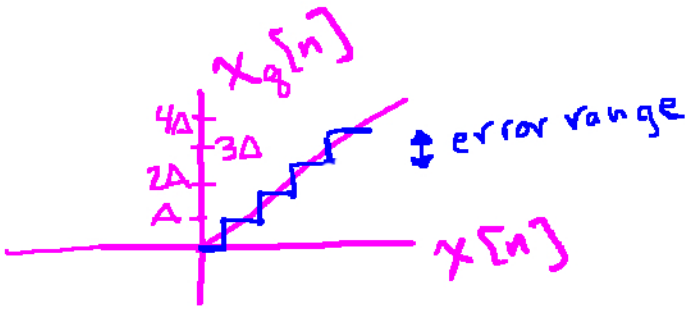
d) The error is independent of the signal being quantized. This implies there does not appear to be any relationship between the error and the signal.

3) This quantization error model is usually accurate if Δ is sufficiently small so that the signal transverses multiple quantization levels between samples and the signal appears random.

4) The ratio of signal power to quantization error power (SQNR) is approximately $6.02b - 1.25$ dB if $b + 1$ bits are used. This implies each additional bit improves the SQNR by 6 dB.

Video Screenshots

Quantization Error Analysis

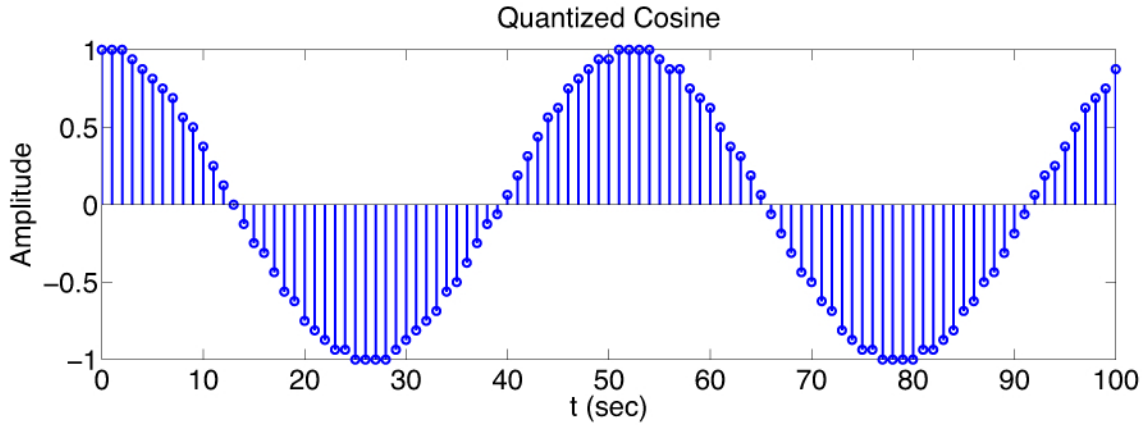


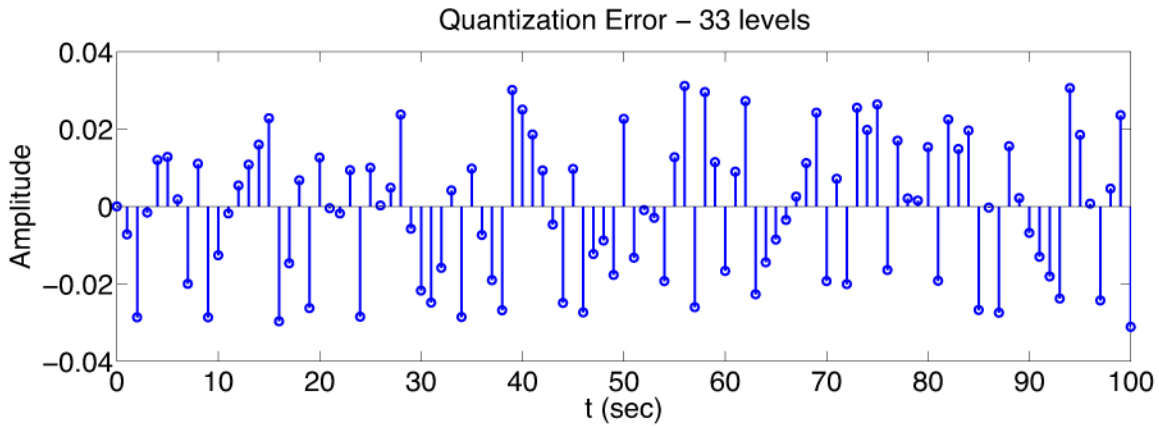
$$e_q[n] = x[n] - x_q[n]$$

$$-\Delta/2 \le e_q[n] \le \Delta/2$$

$e_q[n]$ is complicated or unpredictable

⇒ Statistical characterization





Terminology

1) Stationary – stat. properties do not change with time

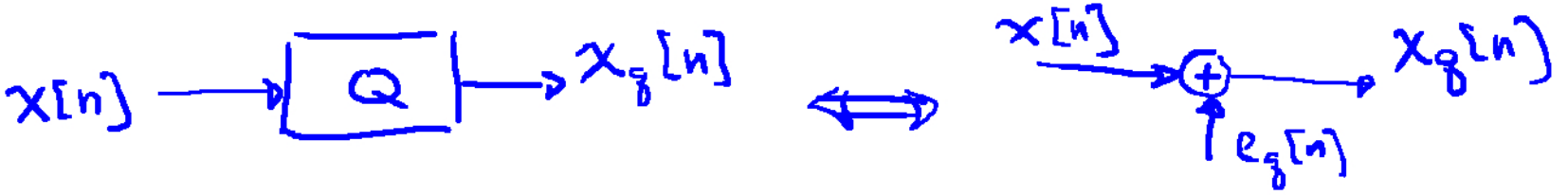
2) Random process – model for a signal with uncertain characteristics

3) Probability distribution – describes how likely it is that we observe the different possible values of a random quantity

4) Uncorrelated – values of signal at any two times appear unrelated

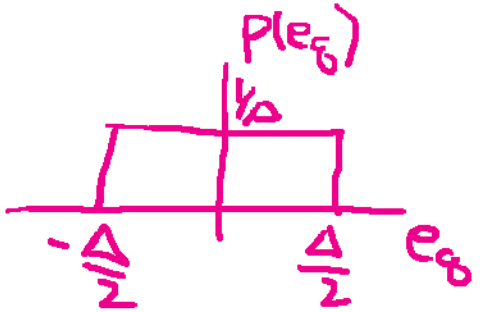
5) Independent – no relationship or pattern of any sort

Quantization Error Model



1) $e_q[n]$ is an observation of a stationary random process

2) $e_q[n]$ is uniformly distributed on $(-\frac{\Delta}{2}, \frac{\Delta}{2})$



3) $e_q[n]$ is uncorrelated

4) $e_q[n]$ is independent of $x[n]$

Reasonable if: Δ is small enough so $x[n]$ traverses several quant levels between samples

$x[n]$ appears random

Signal to Quantization Noise Ratio

$$SQNR = 10\log_{10}\left(\frac{\sigma_x^2}{\sigma_e^2}\right)\ dB$$

σ_x^2 : "power" in $x[n] = E\{x^2[n]\}$

σ_e^2 : "power" in $e_g[n] = E\{e_g^2[n]\}$

$$\sigma_e^2 = \int_{-\frac{\Delta}{2}}^{\Delta/2} e^2 p(e)\,de = \int_{-\frac{\Delta}{2}}^{\Delta/2} e^2\left(\frac{1}{\Delta}\right)de = \frac{e^3}{3\Delta}\Big|_{-\Delta/2}^{\Delta/2} = \frac{\Delta^2}{12}$$

$$SQNR = 10\log_{10}\left(\frac{12\sigma_x^2}{\Delta^2}\right) \qquad \text{but } \Delta = R/2^{b+1}$$

$$SQNR = 10\log_{10}\left(12\sigma_x^2\frac{(2^{b+1})^2}{R^2}\right)$$

$$= 20b\log_{10}(2) + 20\log_{10}(2) + 10\log_{10}(12) - 20\log_{10}\left(\frac{R}{\sigma_x}\right)$$

Let $R = 4\sigma_x - (-4\sigma_x) \Rightarrow \pm 4$ standard deviations

$$SNRQ \simeq 6.02\,b - 1.25$$

6dB per bit!

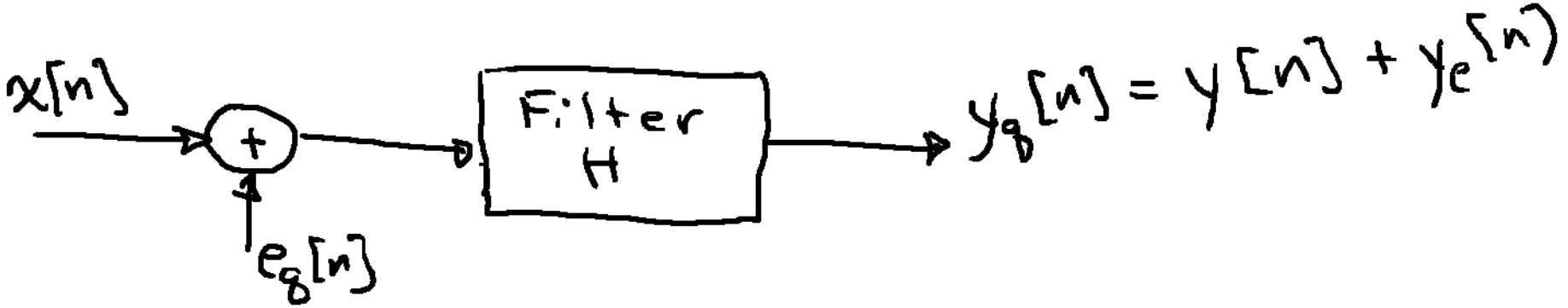
For SNRQ=40

$\Rightarrow b = 7$

so require

8 bits

Quantization Noise and Filtering





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